

Intrinsic Capacity of Chinese Community-Dwelling Older Adults Using WHO Integrated Care for Older People (ICOPE) Framework: Structural Equation Model Analysis

Junjun Sun^{1,2} , Kwanjai Amnatsatsue¹ , Patsri Srisuwan³, Patcharaporn Kerdmongkol¹ , and Dechavudh Nityasuddhi⁴

Abstract

Objective: This study aimed to examine the intrinsic capacity in Chinese older adults and its determinants (sociodemographic, lifestyle-related, environmental, and health-related factors) based on the WHO Integrated Care for Older People (ICOPE) framework.

Methods: A cross-sectional community-based study was conducted among 485 community-dwelling older adults recruited from 4 community health service centers in Xinxiang City, Henan Province between August 2024 and February 2025. Intrinsic capacity was assessed across 5 domains: locomotion, cognition, vitality, psychology, and sensory capacity (hearing and vision). Structural equation model (SEM) was used to examine relationships among intrinsic capacity, sociodemographic factors, lifestyle, family support, social participation, healthcare access, and health conditions.

Results: More than a half of subjects were female (61.2%), married (84.7%), having less than high school education (52%), with a mean age of 73.7 years (SD = 5.5). About 55.9% showed intrinsic capacity impairment, with most impairment of cognition (49.9%), followed by vision (44.9%), hearing (23.7%), locomotion (22.5%), vitality (18.1%), and psychological capacity (13%). The SEM demonstrated good fit and explained 83% of the variance in intrinsic capacity. Significant determinants, ranked by effect size, included dietary habits ($\beta = .544$), age ($\beta = -.303$), pain ($\beta = -.240$), handgrip strength ($\beta = .205$), social participation ($\beta = .181$), access to healthcare services ($\beta = .138$), hospitalization ($\beta = -.10$), and physical exercise ($\beta = .089$). All paths in the model were statistically significant ($p < .05$).

Conclusions: The SEM analysis demonstrated that dietary habits, handgrip strength, social participation, access to healthcare services, and physical exercise were positively associated with intrinsic capacity, while age, pain, and hospitalization showed negative associations. These findings suggest that primary care approaches focusing on multi-domain interventions, particularly health promotion, self-care, and early screening, may be promising areas for future public health initiatives. Community-based programs integrating these modifiable factors should be prioritized to promote intrinsic capacity and functional independence among Chinese community-dwelling older adults.

Keywords

intrinsic capacity, Chinese older adults, ICOPE framework, structural equation model, primary healthcare

Received: March 27, 2025; revised: May 9, 2025; accepted: May 12, 2025

Introduction

Population aging presents unprecedented challenges to global health and social care systems, with the proportion of adults aged 60 years and older projected to increase from 12% to 22% between 2015 and 2050.¹ The rising

socioeconomic burden on healthcare systems was closely linked to increased prevalence of multimorbidity and disability in populations with high life expectancy. However, traditional disease-centered approaches to aging research have proven insufficient, as the absence of disease does not necessarily indicate healthy aging.² In response to this



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons

Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

evolving healthcare landscape, the WHO introduced a groundbreaking model in 2015 that shifts emphasis from disease-centered to function-centered paradigms.³ This model defines healthy aging as “the process of developing and maintaining functional ability that enables well-being in older age.” Central to this model is intrinsic capacity—the composite of all physical and mental capacities that individuals draw upon throughout their life course—encompassing 5 key domains: locomotion, vitality, cognition, psychological well-being, and sensory function.⁴

The WHO has published the Integrated Care for Older People (ICOPE) framework to guide assessing and promoting intrinsic capacity of older adults.⁵ Within ICOPE, the assessment of intrinsic capacity involves in-depth evaluation across 5 domains using standardized measurements. This approach serves as a valuable tool for promoting healthy aging by identifying and addressing declines in intrinsic capacity, thereby facilitating more person-centered and effective interventions. A decline in 1 or more of the 5 domains is considered to constitute intrinsic capacity impairment. A recent meta-analyze has revealed concerning prevalence rates, with 67.8% of community-dwelling older adults globally and 66.0% in China experiencing intrinsic capacity impairment.⁶ This impairment triggers a broad spectrum of adverse outcomes, including functional decline, frailty, mortality, poor quality of life, and so on.⁷ Identifying key determinants that influence intrinsic capacity is therefore essential for developing targeted screening protocols and interventions that mitigate functional decline, preserve independence, and optimize quality of life among older adults.

Recent studies have explored various factors associated with intrinsic capacity among community-dwelling older adults, primarily focusing on sociodemographic characteristics (eg, advanced age and education), lifestyle behaviors (eg, tobacco use and alcohol consumption), and health indicators (eg, comorbidity and handgrip strength).^{8,9} However, a critical limitation persisted: most studies have inadequately addressed environmental influences on intrinsic capacity.¹⁰ The International Classification of Functioning, Disability and Health (ICF) framework highlights environmental contexts as significant contributors to intrinsic capacity, emphasizing the dynamic interaction between these external factors and individual health conditions.¹¹ This consideration is particularly relevant in China, where the “90-7-3 model” indicates approximately 90% of older

adults rely on family-based care and 7% on community-based services, highlighting the critical role of family and community support.¹² Additionally, research has demonstrated that health service access—including geographic accessibility, service availability, and financial affordability—significantly impacted health outcomes in older populations.¹³ Despite their importance, these environmental determinants remain understudied in current research. Compared to general regression analysis, structural equation modeling (SEM) offers distinct advantages¹⁴ by enabling the examination of interdependent relationships among multiple influencing factors and their simultaneous impacts on intrinsic capacity. Furthermore, SEM allows intrinsic capacity to be conceptualized as a latent variable measured through 5 domains, which aligns with the theoretical framework of intrinsic capacity as a holistic construct. Moreover, existing studies have disproportionately focused on either economically advantaged urban centers like Beijing or economically disadvantaged rural areas,^{15,16} with limited evidence from medium-sized Chinese cities. As a third-tier city in central Henan, Xinxiang represents an intermediate level of economic development. Furthermore, its demographic structure closely mirrors Henan Province’s aging pattern, with 17.68% of its 6.17 million permanent residents aged 60 years and above, and 13.04% aged 65 years and above, perfectly aligned with provincial average.¹⁷ Therefore, this study selected Xinxiang City in Henan Province as the research location, filling a critical research gap between China’s economically advantaged urban centers and disadvantaged rural areas.

This study, adopting the WHO ICOPE framework, assessed intrinsic capacity impairment and investigated the relationships among intrinsic capacity, sociodemographic factors, lifestyle behaviors, family support, social participation, healthcare access, and health conditions using SEM. The research was conducted among community-dwelling older adults in Xinxiang City, Henan Province.

Methods

Design and Sample

This cross-sectional study involved older adults from Xinxiang City who were selected through a multi-stage cluster sampling approach. First, all 4 administrative districts of

¹Mahidol University, Bangkok, Thailand

²XinXiang Medical University, Henan, China

³Phramongkutklao Hospital, Bangkok, Thailand

⁴Independent Scholar, Bangkok, Thailand

Corresponding Author:

Kwanjai Amnatsatsue, Department of Public Health Nursing, Faculty of Public Health, Mahidol University, 420/1 Ratchawithi Road., Thung Phaya Thai, Ratchathewi, Bangkok 10400, Thailand.
Email: Kwanjai.amn@mahidol.ac.th

Xinxiang City were included to ensure comprehensive geographic representation. Second, 1 community health service center was randomly selected from each district, resulting in 4 centers for data collection. Finally, we extracted a complete list of elderly residents from each center's electronic health records and used computer-generated random number tables to select participants according to predetermined sample sizes. Selected individuals were contacted by telephone, informed about the study, and provided informed consent prior to participation.

Data collection occurred from September 2024 to February 2025, with face-to-face surveys conducted by trained researchers at either community health centers or participants' homes according to participant preference. Physical assessments were carefully monitored to ensure participant safety, and researchers provided emotional support particularly during sensitive assessments such as the Geriatric Depression Scale. For quality control, data was independently entered by 2 research assistants to ensure consistency in data entry.

Study Participants

Study participants included older adults (≥ 65 years) registered with 4 community health service centers in Xinxiang City from August 2024 to February 2025. Those who met the inclusion criteria were invited to participate in the study. Eligibility required Chinese language proficiency and voluntary consent. Individuals with severe mental disorders, significant language/hearing impairments, or totally dependent living status (Katz ADL ≤ 2) were excluded. Face-to-face questionnaire surveys were conducted, yielding 485 valid responses from 500 distributed questionnaires.

Measures

Intrinsic capacity was evaluated across 5 domains as recommended by the ICOPE: vitality, cognition, locomotion, psychological, and sensory capacities. The vitality domain was assessed using the Short-Form Mini Nutritional Assessment (MNA-SF).¹⁸ Cognitive capacity was measured through the Montreal Cognitive Assessment—Basic (MoCA-B), with scoring criteria adjusted for participants' educational levels.¹⁹ Locomotion was evaluated using the Short Physical Performance Battery (SPPB).²⁰ The psychological domain was assessed with the Geriatric Depression Scale-15 (GDS-15).²¹ Finally, the sensory domain, encompassing both visual and hearing capacities, was evaluated using the E visual acuity chart and Finger rub test with supplementary self-reported questions.^{22,23} Each domain was scored on a scale of 0 to 2 points, where 0 represented severe impairment, 1 indicated mild or moderate impairment, and 2 reflected normal function. The total intrinsic capacity score ranged from 0 to 10 points, with higher scores

indicating better functional capacity. Participants scoring between 0 and 8 were classified as having intrinsic capacity impairment.

Sociodemographic factors included age, sex, educational level, marriage, monthly income, income sufficiency, and living arrangements.

Lifestyle-related factors included physical exercise, sedentary time, dietary habits, sleep habits, alcohol drink, and smoking habits. Physical exercise was assessed through 3 items measuring frequency of vigorous, moderate, and light exercises (scores = 3–12, higher indicating greater exercise). Sedentary time quantified daily minutes spent in activities such as watching TV, reading, using computers, or doing handcrafts. Dietary habits were evaluated using a 10-item questionnaire assessing adherence to recommended intake across food groups (grains, vegetables, fruits, soy products, eggs, meat, fish, dairy, oil, and salt), with total scores ranging from 10 to 40 (higher scores indicating better adherence to dietary guidelines).

Environmental factors included family support, social participation, and access to health services. Family support was assessed using family APGAR questionnaire measuring satisfaction with family functioning, with scores ranging from 0 to 10 (higher scores indicating greater satisfaction).²⁴ Social participation was measured using a scale assessing engagement in diverse activities, with scores ranging from 8 to 40 (higher scores indicating greater participation).²⁵ Healthcare service access was evaluated through 3 items measuring geographic accessibility, service availability, and financial affordability, with scores ranging from 3 to 8 (higher scores indicating better access).¹³

Health-related factors included illness (number of physician-diagnosed conditions), pain presence, pain scale (Visual Analog Scale), sleep quality satisfaction, handgrip strength (kg, measured twice from dominant hand), hospitalization, and falls (number in past 1 year).

Statistical Analysis

Statistical analyses were performed using SPSS 26.0 and Amos 26.0 software. Independent samples *t*-tests were used to compare mean differences in intrinsic capacity between 2 groups. One-way analysis of variance (ANOVA) was employed to assess mean differences in intrinsic capacity among 3 or more groups. Spearman correlation analysis evaluated relationships between continuous independent variables and intrinsic capacity. Effects with $P < .05$ were considered statistically significant.

SEM was employed for path analysis to explore the direct effects of sociodemographic, lifestyle-related, environmental, and health-related factors on older adults' intrinsic capacity. Model fit was evaluated using multiple indices: a χ^2/df ratio < 5 , Root Mean Square Error of Approximation (RMSEA) values ≤ 0.08 , and values ≥ 0.90 for Comparative

Table 1. Sociodemographic Characteristics (N=485).

Variable	Percentage
Age (years)	65-91 (range)
Sex	
Male	188 (38.8)
Female	297 (61.2)
Educational level	
No formal education	37 (7.6)
Primary school	86 (17.7)
Junior high school	132 (27.3)
High school/technical secondary school	109 (22.5)
College and above	121 (24.9)
Marriage	
Married	411 (84.7)
Separated/divorced	65 (13.4)
Widowed	9 (1.9)
Monthly income (CNY)	
<1000	97 (20.0)
1000-1999	33 (6.8)
2000-2999	50 (10.3)
3000-3999	193 (39.8)
4000-4999	64 (13.2)
≥5000	48 (9.9)
Income sufficiency	
Sufficient	22 (4.5)
Just enough, with no extra	251 (51.8)
Insufficient	212 (43.7)
Living arrangements	
Living with both a spouse and children	124 (25.5)
Living with a spouse only	253 (52.2)
Living with children only	62 (12.8)
Living alone	46 (9.5)

Fit Index (CFI), Goodness of Fit Index (GFI), Adjusted Goodness of Fit Index (AGFI), and Tucker–Lewis Index (TLI) were considered indicative of good fit. This article is in accordance with the STROBE guidelines.

Ethical Approval

Ethical approval was obtained from the Xinxiang Medical University Ethics Committee (XYLL-20240279) and the Ethical Review Committee for Human Research, Mahidol University (MUPH2024-081).

Results

The sample comprised 485 older adults aged 65 to 91 years (73.7 ± 5.5), with a majority being female (61.2%) and married (84.7%). Sociodemographic characteristics were shown in Table 1.

Intrinsic capacity scores in the study population ranged from 0 to 10 (mean = 7.97 ± 1.81), with 55.9% of participants ($n=271$) displaying impairment (scores ≤ 8). The

prevalence rates of impairment were 57.1% for sensory capacity ($n=277$), 49.9% for cognition capacity ($n=242$), 22.5% for locomotion capacity ($n=109$), 18.1% for vitality capacity ($n=88$), and 13% for psychological capacity ($n=63$). Regarding sensory capacity specifically, vision impairment was observed in 44.9% of participants ($n=218$), while hearing impairment was present in 23.7% of participants ($n=115$).

Univariate analysis revealed that intrinsic capacity scores were significantly associated with several participant characteristics (Table 2). Educational level, marriage, sleep habits, pain presence, sleep quality satisfaction, and history of hospitalization all demonstrated significant relationships with intrinsic capacity scores (all $P < .05$).

As shown in Table 3, Spearman correlation analysis revealed several significant associations with intrinsic capacity. Dietary habits demonstrated the strongest positive correlation ($r = .525$), followed by weaker positive associations with handgrip ($r = 0.259$), access to health services ($r = .200$), social participation ($r = .154$), and physical exercise ($r = 0.141$; all $P \leq .01$). Among negative correlations, pain scale ($r = -.308$) exhibited moderate inverse relationships, while age ($r = -.296$) and illness ($r = -.159$) showed comparatively weak negative associations (all $P < .001$).

Family support ($r = .078$) and falls ($r = -.085$) showed no significant correlation (all $P > .05$).

In the SEM, intrinsic capacity was operationalized as a latent variable measured by 5 domains. The model was refined using Modification Indices to establish covariance relationships between variables and non-significant paths ($P \geq .05$) were removed to derive the final model (Figure 1). Path coefficient significance was assessed using bootstrap analysis, with results presented in Table 4.

The final SEM demonstrated good fit: $\chi^2/df = 2.752$, GFI = 0.960, AGFI = 0.924, CFI = 0.904, TLI = 0.844, and RMSEA = 0.060 (90% CI = 0.048, 0.073).

The 8 factors in the final model explained 83% of intrinsic capacity variance, highlighting their critical roles in determining intrinsic capacities among older adults. Dietary habits exerted the strongest positive effect ($\beta = .544$, $P = .009$), followed by handgrip strength ($\beta = .205$, $P = .010$), social participation ($\beta = .181$, $P = .013$), access to health services ($\beta = .138$, $P = .009$), and physical exercise ($\beta = 0.089$, $P = .020$), indicating that better dietary habits, stronger grip strength, higher levels of social engagement, greater health-care accessibility, and more frequent physical exercise were associated with enhanced intrinsic capacity among older adults.

Conversely, age ($\beta = -.303$, $P = .020$), pain intensity ($\beta = -.240$, $P = .007$), and recent hospitalization ($\beta = -.100$, $P = .026$) demonstrated negative effects on intrinsic capacity, indicating that advancing age, increased pain levels, and recent hospitalization experiences were associated with diminished intrinsic capacity in older adults.

Table 2. Association Between Categorical Independent Factors and Intrinsic Capacity (n = 485).

Variable	IC scores ($M \pm SD$)	t/F	P-value
Sex			
Male	8.17 ± 1.434	$t = 1.949$	.052
Female	7.84 ± 2.010		
Educational level			
No formal education	7.30 ± 2.379	$F = 2.969$	.019
Primary school	8.02 ± 1.756		
Junior high school	8.30 ± 1.653		
High school/technical secondary school	8.01 ± 1.664		
College and above	7.74 ± 1.896		
Marriage			
Married	8.04 ± 1.740	$t = 2.005$	.046
Separated/divorced/widowed	7.58 ± 2.152		
Monthly income (CNY)			
<1000	7.58 ± 1.600	$F = 1.985$	.079
1000-1999	7.55 ± 1.986		
2000-2999	7.92 ± 1.589		
3000-3999	8.17 ± 1.742		
4000-4999	8.02 ± 2.059		
≥ 5000	8.25 ± 2.139		
Income sufficiency			
Insufficient	7.88 ± 1.934	$F = 0.948$	.388
Just enough, with no extra	8.07 ± 1.686		
Sufficient	7.68 ± 2.033		
Living arrangements			
Not alone	7.96 ± 1.832	$t = -0.378$	.706
Living alone	8.07 ± 1.652		
Sedentary time			
Low sedentary (≤ 120 min/day)	7.78 ± 1.878	$F = 1.732$	.159
Moderate sedentary (121-240 min/day)	8.18 ± 1.679		
High sedentary (241-359 min/day)	7.86 ± 1.894		
Very high sedentary (≥ 360 min/day)	7.82 ± 1.962		
Sleep habits			
Often (5-7 days/week)	8.21 ± 1.574	$F = 9.030$	<.001
Sometimes (3-4 days/week)	7.82 ± 1.966		
Rarely (1-2 days/week)	7.03 ± 2.261		
Not at all (0 days/week)	6.38 ± 2.567		
Alcohol habits			
No	7.93 ± 1.915	$t = -0.791$	.429
Yes	8.07 ± 1.546		
Smoking habits			
No	7.94 ± 1.898	$t = -0.755$	.451
Yes	8.10 ± 1.407		
Pain Presence			
No pain	8.44 ± 1.506	$t = 7.060$	<.001
Have pain	7.32 ± 1.993		
Sleep quality satisfaction			
Completely dissatisfied	6.91 ± 2.308	$F = 7.870$	<.001
Slightly dissatisfied	7.78 ± 1.962		
Slightly satisfied	8.19 ± 1.556		
Completely satisfied	8.22 ± 1.593		
Hospitalization (in the past 1 year)			
No	8.13 ± 1.738	$t = 4.033$	<.001
Yes, once and above	7.32 ± 1.963		

Abbreviation: IC, intrinsic capacity.

Discussion

Our study found an intrinsic capacity impairment prevalence of 55.9% among community-dwelling older adults, comparable to the 55.0% reported in a global meta-analysis,²⁶ but lower than the 67.8% found across 5 countries including China⁶ and the 87.8% reported in Mexico.²⁷ Conversely, our finding exceeds the 39.9% prevalence in the China Comprehensive Geriatric Assessment Study.²⁸ Regarding specific intrinsic capacity domains, the order of

impairment prevalence from highest to lowest—sensory, cognitive, locomotion, vitality, and psychological—matches the sequence reported in a previous study.²⁶ These variations likely reflect our medium-sized city context, addressing a gap between previous studies focused primarily on metropolitan centers or rural areas in China.

The SEM demonstrated good fit and explained 83% of the variance in intrinsic capacity. It validates the comprehensive impact of sociodemographic factors, lifestyle-related factors, environmental factors, and health-related factors on intrinsic capacity, as conceptualized in the ICF framework. Furthermore, our findings support ICOPE's emphasis on comprehensive assessment across multiple domains, as evidenced by the significant pathways from intrinsic capacity to its measurement constructs: sensory capacity, cognitive function (MOCA), physical performance (SPPB), nutrition (MNA), and psychological status (GDS).⁵

Among sociodemographic factors, age emerged as the sole significant variable in the SEM, negatively influencing intrinsic capacity. This finding aligned with previous researches showing that aging both negatively impacted composite intrinsic capacity scores and increased the risk of its domain impairments.^{6,9,27} This trend was confirmed in China, where intrinsic capacity impairment prevalence increased from 28.6% in adults aged 60 to 64 years to 66.6% in those over 80 years.²⁸ Age impacted intrinsic capacity through physiological decline, chronic disease accumulation, and deterioration in cognitive, sensory, and

Table 3. Correlation Between Continues Independent Variables and Intrinsic Capacity (N = 485).

Variables	<i>r</i>	<i>P</i> -value
Sociodemographic factors		
Age	-.296	<.001
Lifestyle-related factors		
Physical exercise	.141	.002
Dietary habits	.525	<.001
Environmental factors		
Social participation	.154	.001
Family support	.078	.088
Access to health services	.200	<.001
Health-related factors		
Illness	-.159	<.001
Pain Scale	-.308	<.001
Handgrip (kg)	.259	<.001
Falls	-.085	.063

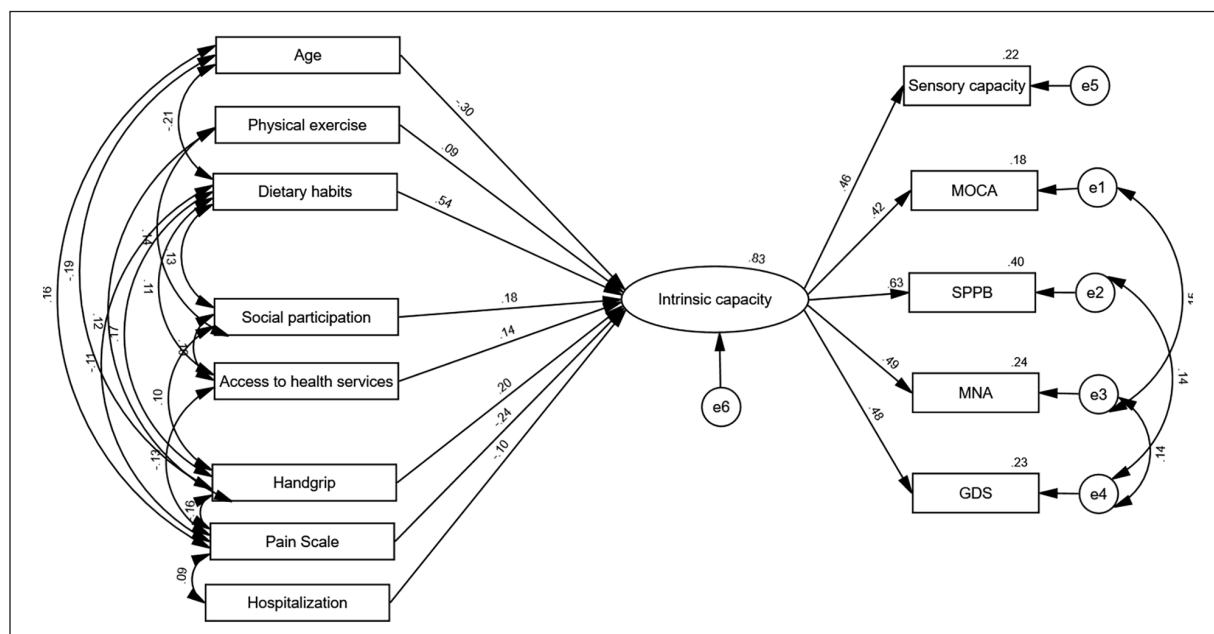


Figure 1. Modified model with standardized parameter estimates.

Abbreviations: GDS, the Geriatric Depression Scale-15; MNA, the Short-Form Mini Nutritional Assessment; MOCA, the Montreal Cognitive Assessment—Basic; SPPB, the Short Physical Performance Battery.

Table 4. Direct Effects of Variables in the Model (N=485).

Path	Standardized estimate	Standard errors	Bootstrapping with 95% CI		P-value
			Lower	Upper	
Age(years)→IC	−0.303	0.054	−0.400	−0.177	.020
Physical exercise→IC	0.089	0.035	0.018	0.147	.020
Dietary habits→IC	0.544	0.050	0.450	0.653	.009
Social participation→IC	0.181	0.042	0.103	0.266	.013
Access to health services→IC	0.138	0.050	0.050	0.248	.009
Handgrip→IC	0.205	0.037	0.138	0.279	.010
Pain Scale→IC	−.240	0.050	−0.334	−0.154	.007
Hospitalization→IC	−0.10	0.047	−0.212	−0.019	.026
IC→Sensory capacity	0.465	0.046	0.377	0.550	.005
IC→MOCA	0.419	0.052	0.322	0.527	.013
IC→SPPB	0.631	0.049	0.527	0.715	.007
IC→MNA	0.489	0.047	0.399	0.583	.004
IC→GDS	0.484	0.047	0.401	0.575	.003

Abbreviation: IC, intrinsic capacity.

psychosocial functions.²⁹ Marriage and education showed significant associations with intrinsic capacity in univariate analysis but not in the final model. Married participants demonstrated better intrinsic capacity, likely due to mutual care and emotional support between spouses.³⁰ Unexpectedly, the univariate analysis revealed a significant but non-linear relationship between educational level and intrinsic capacity: those with no informal education scored lowest, junior high school graduates scored highest, while those with college education and above showed a decline. This pattern may be attributed to moderate education enhancing intrinsic capacity in older adults, while higher education's additional benefits may be moderated by other factors such as occupational stress and lifestyle changes. Further research is needed to better understand these relationships. Sex, income, and living arrangements showed no significant associations.

Regarding lifestyle factors, dietary habits, and physical exercise demonstrated significant positive effects on intrinsic capacity in SEM. Our findings aligned with a previous research demonstrating that “fruits and vegetables” and “protein-rich” dietary patterns improve intrinsic capacity.³¹ Notably, our dietary assessment were based on Chinese Dietary Guidelines focusing on overall dietary balance, offering greater ecological validity than research on specific nutrients. A balanced diet provides comprehensive nutrition, including high-quality protein, essential micronutrients, and dietary fiber—crucial elements for maintaining physiological functions and promoting overall health. Physical exercise benefits likely stem from enhanced cardiorespiratory function, increased muscle strength, elevated brain-derived neurotrophic factors, and improved mental health.³² Sleep quality correlated with superior intrinsic capacity in univariate analysis, while sedentary time, alcohol consumption, and smoking showed no significant

associations. A longitudinal study supports the relationship between sleep and intrinsic capacity, finding that maintaining 7 to 8 h of sleep duration helps maintain optimal intrinsic capacity or delays its decline, while excessive (>10h) and insufficient (<6h) sleep negatively impact intrinsic capacity.³³ This relationship likely exists because healthy sleep habits enhance cognitive function, while poor sleep quality correlates with depression in older adults.³⁴

Environmental factors analysis revealed that social participation and access to healthcare services positively influenced intrinsic capacity in SEM. Enhanced social participation serves as a protective factor against declining intrinsic capacity. This aligns with evidence that social engagement reduces cognitive impairment risk, fulfills belonging needs, and mitigates depression, while social isolation negatively impacts capacity.³⁵ Healthcare accessibility significantly impacts intrinsic capacity, with evidence showing that proximity to healthcare facilities correlates with maintained physical capacity and lower out-of-pocket expenses reduce depression risk.^{36,37} Interestingly, family support showed no significant correlation with intrinsic capacity, reflecting the complex nature of Chinese family relationships where support effects may vary by baseline capacity—potentially beneficial for higher-capacity adults while possibly accelerating decline in lower-capacity individuals through excessive caregiving demands.³⁸

Among health-related factors, pain negatively impacted intrinsic capacity both in SEM and correlation analysis. This finding is consistent with literature demonstrating that pain manifestations correlated with diminished capacity in aging populations, as pain impairs physical functioning, disrupts sleep, and compromises cognitive domains.^{39,40} Handgrip strength demonstrated a significant positive effect, confirming its value as an accessible biomarker

reflecting both musculoskeletal integrity and physiological reserve. Hospitalization history negatively affected intrinsic capacity through mechanisms including bed rest, physical inactivity, medication changes, and psychological stress, with research showing cognitive decline post-hospitalization even after controlling for illness severity.⁴¹ Univariate analysis revealed that participants without multimorbidity demonstrated significantly better intrinsic capacity, suggesting disease management should be central to maintaining capacity in older adults. Unexpectedly, falls showed no significant correlation with intrinsic capacity, contradicting previous research linking falls to decreased physical performance and cognitive decline.⁴² This inconsistency may stem from our assessment methodology, which captured only recent falls without considering their severity or long-term consequences.

Our study has several limitations. First, participants with college education or above constituted 24.9% of our sample, which exceeds the national average of 15.5% for higher education attainment.⁴³ This educational disparity may limit the generalizability of our findings to populations with lower educational levels. Second, our cross-sectional study prevents establishing causal relationships between identified factors and intrinsic capacity; longitudinal studies are needed to validate these associations and track how intrinsic capacity changes over time in response to these determinants. Third, while our SEM examined direct effects of sociodemographic, lifestyle, environmental, and health-related factors on intrinsic capacity, it did not fully address potential interactions among these factors, such as how lifestyle influences might vary depending on health status—an area warranting further investigation.

Conclusion

Our study employed a SEM based on the ICOPE framework to identify key determinants of intrinsic capacity among community-dwelling older adults in China. The SEM analysis revealed that dietary habits, handgrip strength, social participation, healthcare access, and physical exercise positively impact intrinsic capacity, while advanced age, pain intensity, and recent hospitalization negatively affect intrinsic capacity. Primary care providers should implement targeted screening and interventions especially for older adults of advanced age or with recent hospitalizations. This study supports integrated, multi-domain approaches within community settings, such as combining exercise programs with nutritional interventions. These evidence-based insights can inform policy development and resource allocation for community-based healthy aging initiatives that promote functional independence and reduce healthcare burden. Future public health programming should prioritize these modifiable factors to

effectively support intrinsic capacity maintenance among aging populations.

ORCID iDs

Junjun Sun  <https://orcid.org/0000-0002-0086-0647>

Kwanjai Amnatsatsue  <https://orcid.org/0000-0002-6064-3038>

Patcharaporn Kerdmongkol  <https://orcid.org/0009-0001-7784-6405>

Ethical Considerations

Ethical approval was obtained from the Xinxiang Medical University Ethics Committee (XYLL-20240279) and the Ethical Review Committee for Human Research, Mahidol University (MUPH2024-081).

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This thesis was partially supported by the Faculty of Graduate Studies and Graduate Studies of Mahidol University Alumni Association.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

1. World Health Organization. Ageing and health. World Health Organization. 2024. Accessed March 16, 2025. <https://www.who.int/news-room/fact-sheets/detail/ageing-and-health>
2. Zhao J, Chhetri JK, Chang Y, et al. Intrinsic capacity vs. multimorbidity: a function-centered construct predicts disability better than a disease-based approach in a community-dwelling older population cohort. *Front Med*. 2021;8:753295. doi:10.3389/fmed.2021.753295
3. World Health Organization. World Report on Ageing and Health. World Health Organization. 2015. Accessed March 16, 2025. <https://apps.who.int/iris/handle/10665/186463>
4. Cesari M, Araujo de Carvalho I, Amuthavalli Thiagarajan J, et al. Evidence for the domains supporting the construct of intrinsic capacity. *J Gerontol A Biol Sci Med Sci*. 2018;73(12):1653-1660. doi:10.1093/gerona/gly011
5. World Health Organization. Integrated Care for Older People: guidelines on community-level interventions to manage declines in intrinsic capacity. World Health Organization. 2017. Accessed March 16, 2025. <https://www.who.int/publications/i/item/9789241550109>
6. Cao X, Yi X, Chen H, Tian Y, Li S, Zhou J. Prevalence of intrinsic capacity decline among community-dwelling older adults: a systematic review and meta-analysis. *Aging Clin Exp Res*. 2024;36(1):157. doi:10.1007/s40520-024-02816-5
7. Yang Y, Ma G, Wei S, et al. Adverse outcomes of intrinsic capacity in older adults: a scoping review. *Arch Gerontol Geriatr*. 2024;120:105335. doi:10.1016/j.archger.2024.105335

8. Muneera K, Muhammad T, Althaf S. Socio-demographic and lifestyle factors associated with intrinsic capacity among older adults: evidence from India. *BMC Geriatr*. 2022;22(1):851. doi:10.1186/s12877-022-03558-7
9. Jiang X, Chen F, Yang X, et al. Effects of personal and health characteristics on the intrinsic capacity of older adults in the community: a cross-sectional study using the healthy aging framework. *BMC Geriatr*. 2023;23(1):643. doi:10.1186/s12877-023-04362-7
10. Piri AA, Bufali MV, Cappellaro G, Compagni A, Torbica A. Conceptualisation and measurement of healthy ageing: insights from a systematic literature review. *Soc Sci Med*. 2025;374:117933. doi:10.1016/j.socscimed.2025.117933
11. World Health Organization. How to use the ICF: a practical manual for using the International Classification of Functioning, Disability and Health (ICF). World Health Organization. 2013. Accessed March 16, 2025. https://cdn.who.int/media/docs/default-source/classification/icf/drafticf-practicalmanual2.pdf?sfvrsn=8a214b01_4&download=true
12. Nie P, Li Q. Age-friendly community environments and the age trajectories of long-term care dependency among Chinese older adults. *Soc Sci Med*. 2025;365:117614. doi:10.1016/j.socscimed.2024.117614
13. Howdon D, Pasali SS. The impact of ageing on accessibility, affordability and availability of healthcare services in Asia and the Pacific. UN.ESCAP. 2022. Accessed March 16, 2025. <https://hdl.handle.net/20.500.12870/4614>
14. Tomarken AJ, Waller NG. Structural equation modeling: strengths, limitations, and misconceptions. *Annu Rev Clin Psychol*. 2005;1(1):31-65. doi:10.1146/annurev.clinpsy.1.102803.144239
15. Liu S, Kang L, Liu X, et al. Trajectory and correlation of intrinsic capacity and frailty in a Beijing elderly community. *Front Med*. 2021;8:751586. doi:10.3389/fmed.2021.751586
16. Prince MJ, Acosta D, Guerra M, et al. Intrinsic capacity and its associations with incident dependence and mortality in 10/66 Dementia Research Group studies in Latin America, India, and China: a population-based cohort study. *PLoS Med*. 2021;18(9):e1003097. doi:10.1371/journal.pmed.1003097
17. Henan Province Bureau of Statistics. Henan statistical yearbook 2023. China Statistics Press. 2023. Accessed March 16, 2024. <https://tjj.henan.gov.cn/tjfw/tjcbw/tjnj/>
18. Liu H, Jiao J, Zhu M, et al. Nutritional status according to the Short-form Mini Nutritional Assessment (MNA-SF) and clinical characteristics as predictors of length of stay, mortality, and readmissions among older inpatients in China: a national study. *Front Nutr*. 2022;9:815578. doi:10.3389/fnut.2022.815578
19. Huang L, Chen KL, Lin BY, et al. Chinese version of Montreal Cognitive Assessment Basic for discrimination among different severities of Alzheimer's disease. *Neuropsych Dis Treat*. 2018;14:2133-2140. doi:10.2147/NDT.S174293
20. Li W, Rao Z, Fu Y, et al. Value of the Short Physical Performance Battery (SPPB) in predicting fall and fall-induced injury among old Chinese adults. *BMC Geriatr*. 2023;23(1):574. doi:10.1186/s12877-023-04290-6
21. Su H, Zhou Y, Sun Y, Cai Y. The relationship between depression and subjective cognitive decline in older adults of China: the mediating role of general self-efficacy. *Psychol Health Med*. 2023;28(4):1057-1067. doi:10.1080/13548506.2022.2125165
22. Jin J. Screening for impaired visual acuity in older adults. *JAMA*. 2022;327(21):2158. doi:10.1001/jama.2022.7534
23. Strawbridge WJ, Wallhagen MI. Simple tests compare well with a hand-held audiometer for hearing loss screening in primary care. *J Am Geriatr Soc*. 2017;65(10):2282-2284. doi:10.1111/jgs.15044
24. Smilkstein G. The family APGAR: a proposal for a family function test and its use by physicians. *J Fam Pract*. 1978;6(6):1231-1239.
25. Wei X, Wu J. The impact of social participation of Chinese older adults on the risk of death. *South Popul*. 2015;30(2):13. doi:10.3969/j.issn.1004-1613.2015.02.006
26. Jayaraj V, Gnanasekaran S, Yazhini VB, et al. Estimating the prevalence of intrinsic capacity decline: a systematic review and meta-analysis using WHO's integrated care of older people (ICOPE) screening tool. *Arch Gerontol Geriatr Plus*. 2024;1(3):100032. doi:10.1016/j.aggp.2024.100032
27. Gutiérrez-Robledo LM, García-Chanes RE, Pérez-Zepeda MU. Screening intrinsic capacity and its epidemiological characterization: a secondary analysis of the Mexican Health and Aging Study. *Rev Panam Salud Publica*. 2021;45:e121. doi:10.26633/RPSP.2021.121
28. Ma L, Chhetri JK, Zhang L, Sun F, Li Y, Tang Z. Cross-sectional study examining the status of intrinsic capacity decline in community-dwelling older adults in China: prevalence, associated factors and implications for clinical care. *BMJ Open*. 2021;11(1):e043062. doi:10.1136/bmjopen-2020-043062
29. Tenchov R, Sasso JM, Wang X, Zhou QA. Aging hallmarks and progression and age-related diseases: a landscape view of research advancement. *ACS Chem Neurosci*. 2024;15(1):1-30. doi:10.1021/acscchemneuro.3c00531
30. Ng YT, Huo M, Han SH, Birditt KS, Fingerman KL. Older adult's marital status, conversation frequency, and well-being in everyday life. *J Gerontol B Psychol Sci Soc Sci*. 2022;77(3):499-512. doi:10.1093/geronb/gbab112
31. Huang CH, Okada K, Matsushita E, et al. Dietary patterns and intrinsic capacity among community-dwelling older adults: a 3-year prospective cohort study. *Eur J Nutr*. 2021;60(6):3303-3313. doi:10.1007/s00394-021-02505-3
32. Mahindru A, Patil P, Agrawal V. Role of physical activity on mental health and well-being: a review. *Cureus*. 2023;15(1):e33475. doi:10.7759/cureus.33475
33. Chen XL, Li J, Sun SN, et al. Validation of intrinsic capacity and healthy sleep pattern in middle-aged and older adults: a longitudinal Chinese study assessing healthy ageing. *J Nutr Health Aging*. 2024;28(11):100365. doi:10.1016/j.jnha.2024.100365
34. Liao H, Liao S, Gao YJ, Mu JP, Wang X, Chen DS. Correlation between sleep time, sleep quality, and emotional and cognitive function in the elderly. *Biomed Res Int*. 2022;9709536. doi:10.1155/2022/9709536
35. Zhao Y, Chen Y, Xiao LD, et al. Intrinsic capacity trajectories, predictors and associations with care dependence in community-dwelling older adults: a social determinant of health perspective. *Geriatr Nurs*. 2024;56:46-54. doi:10.1016/j.gerinurse.2023.12.022

36. Bialowolski P, Weziak-Bialowolska D, Lee MT, Chen Y, VanderWeele TJ, McNeely E. The role of financial conditions for physical and mental health. Evidence from a longitudinal survey and insurance claims data. *Soc Sci Med*. 2021;281:114041. doi:10.1016/j.socscimed.2021.114041
37. Machón M, Vrotsou K, Larrañaga I, Vergara I. Proximity to facilities and its association with the health-related habits of functionally independent older adults. *Int J Environ Res Public Health*. 2020;17(22):8677. doi:10.3390/ijerph17228677
38. Liu M, Chang Y, Zhao S, et al. The effect of the interaction between intrinsic capacity and social support on the trajectories of activities of daily living in older adults. *Geriatr Nurs*. 2024;60:231-240. doi:10.1016/j.gerinurse.2024.09.007
39. Van Der Leeuw G, Eggermont LH, Shi L, et al. Pain and cognitive function among older adults living in the community. *J Gerontol A Biol Sci Med Sci*. 2016;71(3):398-405. doi:10.1093/gerona/glv166
40. Yuan Q, Yue X, Wang M, et al. Association between pain, sleep and intrinsic capacity in Chinese older adults: evidence from CHARLS. *J Nutr Health Aging*. 2025;29(3):100466. doi:10.1016/j.jnha.2024.100466
41. Wilson RS, Hebert LE, Scherr PA, Dong X, Leurgans SE, Evans DA. Cognitive decline after hospitalization in a community population of older persons. *Neurology*. 2012;78(13):950-956. doi:10.1212/WNL.0b013e31824d5894
42. Trevisan C, Ripamonti E, Grande G, et al. The association between injurious falls and older adults' cognitive function: the role of depressive mood and physical performance. *J Gerontol A Biol Sci Med Sci*. 2021;76(9):1699-1706. doi:10.1093/gerona/glab061
43. Akimov AV, Gemueva KA, Semenova NK. The seventh population census in the PRC: results and prospects of the country's demographic development. *Her Russ Acad Sci*. 2021;91(6):724-735. doi:10.1134/S1019331621060083